Does a coach's rule pick the better of his level's two natural moves?

Four rules for the quiet moments of a game, as tie-breakers between the two moves Maia-2 rates most likely, in 90,000 Lichess games refereed by Stockfish

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Abstract

A coach's rules for the quiet moments of a game are not meant to choose a move alone: his student names a few normal moves, and the rule says which of them to play. This study asks four such rules, from a titled player's review of a club player's rapid games, only to break that tie. In quiet positions from 90,000 Lichess games in the player's band and the one above, where the two moves Maia-2 rates most likely at the mover's band are one that follows a rule and one that breaks it, is the one that follows sound more often than Maia-2's own order accounts for? The rule, fixed before any position was scored, first asks a planted control to show that the design sees a rule known to hold, and it does: a move that leaves a piece to be taken for a gain loses the tie by 29.1 points on the train months and 28.5 held out. Two rules then break the tie, their intervals above 0 on both periods: bishop over knight, by 5.6 and 4.0 points, and pawns off the bishop's colour, by 3.4 and 4.0. Keeping the tension does no better than Maia-2's order, and trading by development breaks the tie on the train months only. Neither winner overrules a first choice. Where Maia-2 rates the move that breaks the rule first, that move is still sound more often than the rule's, by 17 and 10 points held out, and where only one of the two is sound, bishop over knight's move is that one 54% of the time held out, not the coach's four games in five. The two rules choose between two moves that look about equally good; they are no reason to play the one that looks worse.

1 The question

A titled player reviewing the rapid games of an adult player rated about 1100 gives rules for the quiet moments of a game, when nothing is hanging and no tactic is on the board: keep the tension unless taking gains something; avoid even trades when ahead in development and look for them when behind; keep the pawns off the bishop's colour; and prefer the bishop to the knight, the better trade, he says, about four games in five.[1] His fifth rule, the central break while the other king is still in the centre, was tested in an earlier study and showed no signal.

Two earlier studies asked a rule of thumb to choose the move: among the legal moves that show the rule, the one Maia-2 rates most likely at the mover's band. The rule's move, one his level plays less often, was sound less often than the move his level would naturally play, before the tactics were checked and after. The coach does not use his rules that way. The student names a few normal moves, and the rule says which of them to play. So this study asks a rule only to break a tie. In quiet positions where the two moves Maia-2 rates most likely at his level are one that follows the rule and one that breaks it, is the one that follows it sound more often, beyond what Maia-2's own order says?

[1] `https://www.youtube.com/watch?v=PcGDehRSqTM`

If a rule breaks the tie, a lesson can teach it as the way to choose between two natural moves, and Game Review can name it where he played the move that breaks it and his other natural move followed it. If it does no better than Maia-2's order, it explains a move but does not choose between his candidates. If it misleads, a lesson that teaches it should say where it fails.

2 How it was measured

The rules. Four rules, read as six units (Table 1). A unit has a situation and reads each legal move: the move *follows* the unit's rule, *breaks* it, or neither. A rule of two units reads two mirrored halves, so that a liking for quiet moves over trades in general, or for trades over quiet moves, cancels between them. Every reading is the app's knowledge service at commit `ffee8dbb16`. A *quiet move* is not a capture and gives no check. An *even trade* is a capture that gives no check and that the app's capture war scores 0: what it takes, less the opponent's best continuation on its square. A *lever* is a pawn of the mover's that attacks a pawn of the opponent's. Development has no reading in the app, so it is defined here: a side's undeveloped count is its knights and bishops on their starting squares, plus one when its king stands on its starting square, and the mover's lead is the opponent's count less the mover's. Material is level when the mover's points are within 1 of the opponent's, which keeps out the rules about trading when ahead or behind in material.

The positions. Lichess rated games at 10 minutes with no increment, his own time control, that ended normally or on time, with at least 30 plies, both players rated 1250 to 1649: the first 30,000 such games of each month, July and August 2026 to train on and September 2026 held out. A position enters from ply 8 on when the move just played captured nothing, the mover is not in check and has two legal moves or more, and the app gives neither side a mate in one or a winning capture. M_1 and M_2 are the two moves Maia-2, the app's, rates most likely at the mover's band, 1250 to 1449 (his) or the band above. A position *splits* for a unit when it is in the unit's situation and one of M_1 and M_2 follows the unit's rule and the other breaks it: F is the move that follows and B the move that breaks. It is *follow-first* when M_1 is F and *break-first* when M_1 is B. Each game gives a unit at most one position, drawn at random among its splits, and each unit keeps the first 3,000 a month.

The referee. With seven men or fewer the app's tablebase probe decides, and a move is *sound* when it keeps the mover's outcome. Otherwise Stockfish, through the app's engine service at its grading budget of 250,000 nodes with the hash cleared for each position, searches the best and then scores F and B, and a move is sound when it gives up no more than 3 win% against that best. The move played is recorded and not scored.

The run. The rule, every script, the app's code and its installed packages were frozen by hash before any position was scored, and each stage of the run checked the freeze before it started. Once, between two stages, the check found cache files that another checkout's test runner, sharing the run's installed packages on the same machine, had written among them. Only that test runner reads them; once they were moved aside the freeze held again, and the run resumed from its checkpoints.

The measures. The *lift* is F's share sound less B's, paired by position, in points. The move Maia-2 rates first is likely to be sound more often than its second, which adds to the lift in the

Unit	Situation	Follows	Breaks
Tension	the mover has a lever whose capture is an even trade	a quiet move that moves no lever's pawn	a lever's capture that is an even trade
Ahead in development	not an endgame, material level, an even trade of a piece on, the mover 2 or more ahead in development	a quiet move	an even trade of a piece
Behind in development	the same, the mover 2 or more behind	an even trade of a piece	a quiet move
Bishop for knight	the mover can give a bishop for a knight in an even trade, and cannot take a bishop with a knight so	a quiet move	the bishop taking the knight
Knight for bishop	the mover can take a bishop with a knight in an even trade, and cannot give a bishop for a knight so	the knight taking the bishop	a quiet move
Bishop's colour	the mover has one bishop, and quiet pawn moves to squares of both colours	a quiet pawn move to a square of the other colour	a quiet pawn move to a square of the bishop's colour
Level development	as ahead in development, with development level	a quiet move	an even trade of a piece
The control	any position where a rule's unit is in its situation	a quiet move after which the opponent has no winning capture	a quiet piece move that leaves the piece to be taken for a gain

Table 1: The units. The six above the line read the four rules: keep the tension and pawns off the bishop's colour are a unit each, trades and development reads its two halves, ahead and behind in development, and bishop over knight its two, bishop for knight and knight for bishop. Level development is a reference, and no rule: it shows how a quiet move fares against an even trade when development says nothing. The control is planted: a move that leaves its piece to be taken for a gain is known to be the worse of two far more often than not.

follow-first positions and takes from it in the break-first ones. The *balanced lift* is the mean of the two orders' lifts, which cancels that advantage when it is the same in both orders, so a rule that carries no information reads about 0 whatever share of its splits come in each order. A rule's lift is its unit's balanced lift, or the mean of its two units' on every draw. *When only one is sound* is the share of the positions with exactly one of F and B sound where it is F, the mean of the two orders' shares: a rule with no information reads 50%, and the coach's four games in five would read 80%. *Against the first choice* is F's share sound less M_1's: what playing the rule's move in every split would gain over playing Maia-2's first choice.

The rule, fixed before any position was scored. Intervals are 95%, from 2,000 bootstrap draws over games (seed 20261021), within each period, every unit's measures on a draw using the positions of the games drawn. The control comes first: its balanced lift must have its interval above 0 on both periods, with enough data, or no rule receives a verdict. Then a rule *breaks the tie* when its lift's interval lies above 0 on the train months and on the held-out month, *misleads* when it lies below 0 on both, and *does no better than Maia-2's order* otherwise. A rule whose units have fewer than 300 held-out positions, or fewer than 100 in either order, has not enough data. Beside the verdict, and changing none, a rule is *worth switching to* when its lift in the break-first positions

has its interval above 0 on both periods: there his level's first move breaks the rule and its second follows it, so the rule is a reason to play the second.

3 What was scored

The scan read the first 30,000 games of each month, 90,000 in all. 2,055,801 positions entered it, and Maia-2 read the 837,954 of them that stand in some unit's situation, and no other. The draw and the cap kept 54,353 positions, from 24,121 games on the train months and 12,043 held out; a position can serve several units, so the units read 59,616 in all. Tension, bishop for knight, knight for bishop and the bishop's colour filled the cap of 3,000 every month (Table 2). The rarest split is being ahead in development: 1,196 positions held out, 329 of them follow-first. So every unit passes the rule's counts of 300 held-out positions and 100 in each order.

	In situation, held out	Split, held out	Kept, train	Kept, held out	Follow-first	Break-first
Tension	49,779	22,180	6,000	3,000	960	2,040
Ahead in development	3,359	1,454	2,462	1,196	329	867
Behind in development	6,651	2,888	4,879	2,383	1,565	818
Bishop for knight	48,681	13,805	6,000	3,000	1,518	1,482
Knight for bishop	8,975	6,306	6,000	3,000	2,308	692
Bishop's colour	189,254	7,644	6,000	3,000	1,501	1,499
Level development	60,057	28,609	6,000	3,000	1,012	1,988
The control	267,036	1,442	2,436	1,260	799	461

Table 2: Where each unit comes up. In situation: the held-out quiet positions in the unit's situation. Split: those where one of the two natural moves follows the unit's rule and the other breaks it. Kept: at most one position a game, drawn at random among its splits, and at most 3,000 a month. The last two columns divide the held-out kept positions by the move Maia-2 rates first: the one that follows, or the one that breaks.

4 The control passes

The control's balanced lift is 29.1 points (26.3 to 31.8) over 2,436 positions on the train months and 28.5 (24.4 to 32.4) over 1,260 held out, both intervals above 0: the design sees a rule known to hold, and the rules receive verdicts. Where only one of its two moves is sound, it is the move that leaves nothing to be taken 76% of the time on the train months and 75% held out. The control holds against Maia-2's order as well: where Maia-2 rates the move that leaves a piece to be taken first, its second move is sound more often (Table 3). Every rule's interval lies below the control's on both periods, so what follows is about rules that tip a choice, not rules that decide one.

5 Two rules break the tie

Bishop over knight breaks the tie by 5.6 points (4.3 to 6.9) on the train months and 4.0 (2.1 to 5.9) held out, and pawns off the bishop's colour by 3.4 (1.8 to 5.0) and 4.0 (1.7 to 6.1) (Table 3). Keeping the tension does no better than Maia-2's order, at −0.4 points (−2.2 to 1.5) and −1.5

(−4.1 to 1.1). Trading by development breaks the tie on the train months, by 4.5 points (2.6 to 6.3), but not held out, at 1.1 (−1.8 to 3.9); the periods disagree, so by the rule it does no better than Maia-2's order either.

Rule	Balanced lift, train	Balanced lift, held out	Break-first lift, held out	Verdict
Keep the tension	−0.4 (−2.2, 1.5)	−1.5 (−4.1, 1.1)	−17.2 (−20.0, −14.3)	does no better than Maia-2's order
Trades and development	4.5 (2.6, 6.3)	1.1 (−1.8, 3.9)	−16.1 (−19.4, −12.7)	does no better than Maia-2's order
Bishop over knight	5.6 (4.3, 6.9)	4.0 (2.1, 5.9)	−16.6 (−19.6, −13.5)	breaks the tie
Pawns off the bishop's colour	3.4 (1.8, 5.0)	4.0 (1.7, 6.1)	−10.3 (−13.4, −7.0)	breaks the tie
The control	29.1 (26.3, 31.8)	28.5 (24.4, 32.4)	16.3 (9.5, 22.9)	passes

Table 3: The verdicts. The balanced lift is the mean, over the two orders, of the share of positions where the move that follows the rule is sound less the share where the move that breaks it is, in points, with its 95% interval from 2,000 draws over games; a rule of two units takes the mean of its units' lifts on every draw. The break-first lift is the same difference where Maia-2 rates the move that breaks the rule first: a rule is worth switching to when its interval lies above 0 on both periods, and none does.

6 A tie-breaker, not a reason to switch

The balanced lift is half the difference between two leads: how far Maia-2's first choice leads its second in soundness where the first follows the rule, and how far it leads where the first breaks it. For the two winners the first choice leads by more where it follows the rule, which is the rule's information, but it leads in both orders, on both periods (Table 4). So neither winner is worth switching to, and no rule is: every rule's break-first lift lies below 0 on both periods. Where Maia-2 rates the move that breaks the rule first, that move is sound more often than the second, the rule's, by 17 points for bishop over knight and 10 for the bishop's colour held out.

The other measures say the same. Where only one of the two moves is sound, it is the rule's move 55% of the time on the train months and 54% held out for bishop over knight, and 54% and 54% for the bishop's colour: above the 50% of a rule with no information, and below the 80% that the coach's four games in five would read. And playing the rule's move in every split, in place of Maia-2's first choice, costs 5 to 12 points of soundness held out across the four rules, every interval below 0 on both periods.

7 The halves

A rule of two units is read as the mean of its halves, because a lean towards quiet moves, or towards trades, lifts one half and sinks the other by as much. The halves are reported apart and decide nothing (Table 4).

Bishop over knight's halves differ. Keeping the bishop rather than giving it for a knight lifts by 9.5 points (7.7 to 11.1) on the train months and 8.1 (5.8 to 10.7) held out, while taking a bishop with a knight reads 1.7 (−0.4 to 3.8) and −0.1 (−3.1 to 2.9), across 0 on both. That fits a rule that holds

only for keeping the bishop, and as well one that holds both ways beside a lean towards the quiet move, which the mean cancels; the halves cannot tell the two apart. The trade units side with the rule both ways. In each of them, on both periods, a trade that takes a bishop fares better against the quiet move than one that takes a knight, and when behind in development, the trade lifts by 21.5 points held out where it takes a bishop and loses by 8.6 where it takes a knight, both intervals clear of 0 on both periods.

Of trading by development's halves, trading when behind lifts by 5.1 points (3.0 to 7.0) and 3.8 (1.0 to 6.6), and keeping the pieces on when ahead by 3.9 (0.8 to 6.9) on the train months but −1.5 (−6.4 to 3.1) held out. The reference, a quiet move against an even trade of a piece with development level, reads 0.7 (−1.1 to 2.5) and −1.1 (−3.7 to 1.5): with development level, neither the quiet move nor the trade fares better than Maia-2's order says.

	Held-out positions	F sound	B sound	Lift, follow-first	Lift, break-first	Balanced lift	F when only one is sound
Tension	3,000	57	65	14.1 (9.7, 18.3)	−17.2 (−20.0, −14.3)	−1.5 (−4.1, 1.1)	49
Ahead in development	1,196	52	59	10.9 (2.9, 19.4)	−14.0 (−18.6, −9.1)	−1.5 (−6.4, 3.1)	48
Behind in development	2,383	67	56	25.8 (22.5, 29.1)	−18.2 (−23.0, −13.6)	3.8 (1.0, 6.6)	53
Bishop for knight	3,000	63	54	27.1 (23.6, 30.5)	−10.8 (−14.2, −7.1)	8.1 (5.8, 10.7)	57
Knight for bishop	3,000	66	54	22.2 (19.3, 25.0)	−22.4 (−27.7, −17.3)	−0.1 (−3.1, 2.9)	50
Bishop's colour	3,000	67	63	18.3 (15.0, 21.5)	−10.3 (−13.4, −7.0)	4.0 (1.7, 6.1)	54
Level development	3,000	55	63	19.2 (15.0, 23.4)	−21.4 (−24.6, −18.3)	−1.1 (−3.7, 1.5)	49
The control	1,260	62	30	40.7 (36.0, 45.0)	16.3 (9.5, 22.9)	28.5 (24.4, 32.4)	75

Table 4: Each unit apart, held out. F and B sound: the share of positions where the move that follows and the move that breaks the unit's rule is sound, in percent. The lifts are F's share less B's in points, in each order and as their mean, with 95% intervals. F when only one is sound: among the positions where exactly one of the two is sound, the share where it is F, as the mean of the two orders' shares, in percent.

8 Where the two rules count

Close calls. A close call is a position where Maia-2 rates its second move at least half as likely as its first: 41% to 56% of a rule unit's held-out positions (Table 5). There a tie-breaker matters most, since his level plays either move. In close calls, keeping the bishop where the first choice gives it for a knight is sounder than that first choice by 4.2 points (0.8 to 7.6) on the train months, and by 2.4 (−2.2 to 7.3) held out, where the interval spans 0. A pawn kept off the bishop's colour where the first choice puts one on it reads −1.5 (−4.6 to 1.4) and −2.1 (−6.2 to 2.2), about as sound as that first choice. Where the first choice is clear, it wins: there, following the rule against it costs 26 points for keeping the bishop and 20 for the bishop's colour held out, below 0 on both periods.

A quiet move also fares better against an even trade in close calls when development is level, by 4.9 and 3.8 points with both intervals above 0, and so does keeping the tension on the train months, by 4.3 points, though not held out, at 3.2. Part of a quiet move's lift in a close call may be that lean, which this study does not separate. The bishop's colour compares two quiet pawn moves, so no such lean touches it.

Phase and band. Keeping the bishop lifts most in the opening, by 10.2 points held out against 7.0 in the middlegame, where it still lifts on both periods, and more in his band than in the band

	Close calls			Clear first choice		
	Positions	Balanced lift	Break-first lift	Positions	Balanced lift	Break-first lift
Tension	1,414	3.2 (−0.4, 6.8)	−2.8 (−7.3, 1.7)	1,586	−2.5 (−6.3, 1.2)	−26.9 (−30.6, −23.0)
Ahead in development	563	2.1 (−4.4, 8.2)	−0.3 (−7.8, 7.4)	633	0.5 (−7.0, 8.0)	−22.9 (−29.2, −16.3)
Behind in development	1,167	−1.8 (−6.0, 2.2)	−12.6 (−18.6, −6.7)	1,216	5.0 (0.4, 9.3)	−27.8 (−35.8, −19.9)
Bishop for knight	1,673	9.8 (6.4, 13.1)	2.4 (−2.2, 7.3)	1,327	7.2 (3.7, 10.8)	−26.2 (−31.0, −21.1)
Knight for bishop	1,219	−6.2 (−10.3, −2.2)	−17.4 (−23.9, −11.0)	1,781	−1.7 (−6.5, 3.2)	−33.6 (−42.7, −24.8)
Bishop's colour	1,659	3.4 (0.3, 6.5)	−2.1 (−6.2, 2.2)	1,341	5.2 (1.9, 8.5)	−19.9 (−24.8, −14.9)
Level development	1,409	3.8 (0.2, 7.5)	−3.0 (−7.9, 2.1)	1,591	0.7 (−3.1, 4.6)	−32.8 (−36.8, −29.1)
The control	718	26.5 (21.4, 31.6)	21.4 (13.8, 28.7)	542	26.1 (19.0, 33.0)	3.1 (−10.0, 16.2)

Table 5: Close calls against a clear first choice, held out, in points with 95% intervals. A close call: Maia-2's probability of its second move is at least half its probability of its first. The break-first lift is F's soundness less B's where Maia-2 rates B first. The splits describe and decide nothing.

above, by 11.5 against 4.9 (Table 6). Taking a bishop with a knight lifts in the opening, by 8.5, and misleads in the middlegame, at −4.4, both clear of 0 on both periods. The bishop's colour lifts in the middlegame and the endgame, by 5.5 and 6.0 points held out, above 0 on both periods, and not in the opening, at −2.7 with its interval across 0 on both; it lifts in both bands.

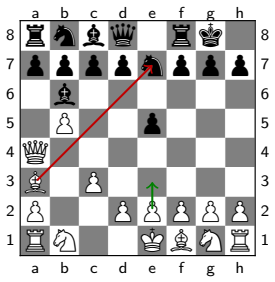
	1250 to 1449	1450 to 1649	Opening	Middle-game	Endgame
Tension	−2 (1,299)	−1 (1,701)	−2 (1,304)	−3 (1,261)	5 (435)
Ahead in development	−1 (538)	−2 (658)	−4 (763)	2 (433)	–
Behind in development	0 (1,083)	7 (1,300)	2 (1,598)	7 (785)	–
Bishop for knight	12 (1,450)	5 (1,550)	10 (1,354)	7 (1,509)	−3 (137)
Knight for bishop	−2 (1,275)	1 (1,725)	8 (1,061)	−4 (1,715)	−6 (224)
Bishop's colour	4 (1,364)	4 (1,636)	−3 (572)	6 (1,704)	6 (724)
Level development	1 (1,332)	−3 (1,668)	0 (1,450)	−3 (1,550)	–
The control	30 (572)	27 (688)	23 (211)	30 (930)	24 (119)

Table 6: The balanced lift by the mover's band and by phase, held out, in points, with the cell's positions in brackets; a dash is a cell with none. The bands are the mover's Lichess rating; the phase is the app's divider's. The splits describe and decide nothing; every cell's interval is in `results/rules.json`.

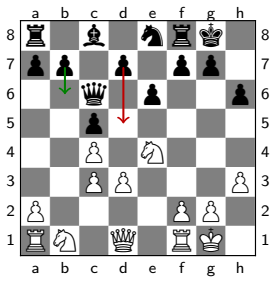
9 Examples

The rule draws six held-out examples a unit, at random among the positions where exactly one of the two moves is sound. For keeping the bishop, the rule's move is the sound one in three of its six and the other move in three; for the bishop's colour, in four and two. Figure 1 shows three.

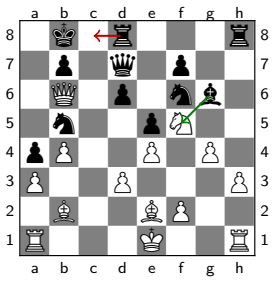
In the first, White, in the band above his, can give a bishop for a knight with 6. ♗xe7, which Maia-2 rates first, ahead of the quiet 6. e3. By the referee, White's win% is 43 after the quiet move and 35 after the trade; the referee's best is 6. ♘f3, and the game went 6. c4. In the second, Black, in his band, can put a pawn on the bishop's colour with 14... d5, Maia-2's first choice, or keep it off with 14... b6, which was played and is the referee's best: Black's win% is 69 after it and 64 after 14... d5. The third is one where the rule is wrong. Black, in the band above his, played the quiet 24... ♖c8, Maia-2's first choice, but giving the bishop for the knight with 24... ♗xf5 is the referee's best: Black's win% is 41 after the trade and 24 after the quiet move.



(a) Bishop for knight, break-first: 6. e3 is sound, 6. ♗xe7 is not.



(b) The bishop's colour, break-first: 14. . . b6 is sound, 14. . . d5 is not.



(c) Bishop for knight, follow-first: 24. . . ♗xf5 is sound, 24. . . ♖c8 is not.

Figure 1: Three of the held-out examples, drawn at random among the positions where exactly one of the two natural moves is sound. Green: the sound move; red: the unsound one.

10 What it means for the app

Two of the coach's four rules carry information that his level's own order of moves lacks, and they read best as he gives them: as the way to choose between two normal moves. A lesson can teach them as tie-breakers. When two moves look about equally good and one gives the bishop for a knight, play the other; when two pawn moves look about equally good, push the one that goes to the other colour from the bishop. Neither is a reason to overrule a move that looks better: where his level's first choice breaks the rule, that first choice is still sound more often than the rule's move, and only in a close call is the rule's move about as sound. The bishop's colour counts in the middlegame as in the endgame, where the app's catalogue already teaches it, and not in the opening.

Keeping the tension and trading by development did not choose between his two natural moves, so a lesson that teaches them explains a move, but should not offer them as the way to choose one.

11 Caveats

- **Maia-2's order is the yardstick.** The balanced lift assumes the first choice's lead over the second is about the same in the two orders. A rule whose follow-first positions differ in kind from its break-first ones could lean either way, so each order's lift is reported, and the control shows the design sees a rule known to hold.
- **Halves are not tests.** Only a rule's mean was tested. A half's lift can carry a lean towards quiet moves or towards trades, so the halves describe where a rule's lift sits, not which half holds.
- **Many splits.** The splits are many, and some cells will read clear of 0 by chance; they describe and decide nothing.
- **Not his games.** The positions are Lichess players' in his band and the band above; his own games are too few for a test like this.
- **Two candidates.** The coach's student names a few normal moves. The test takes the two Maia-2 rates most likely at the mover's band, and only where one follows the rule and the other breaks it.
- **Quiet by board code.** No mate in one and no winning capture for either side is the app's reading of a quiet moment. A deeper tactic, a fork two moves away, still counts as quiet.

- **Even by the capture war.** A trade the app's capture war calls even can still lose to a tactic it does not see; that is the referee's to find, and part of what a rule's move is judged on.
- **Development by the starting squares.** The app has no reading of development, so the count of undeveloped minor pieces and the king on its square is this study's. A rook not yet connected or a queen out early does not count.
- **One move for a plan.** A rule is tested on the next move, not as a plan over several, and a sound move can still be the second-best plan.
- **Sound within a window.** A move is sound within 3 win% of the best; a rule whose gain is smaller than that, or comes later than the referee's search sees, reads as none.